

SCCM 2509 Lab

Installation & Troubleshooting — Deep-Dive Report

Site Server: **SCCM01.corp.lab** · SQL: **SCCM-DB.corp.lab** · Domain: **corp.lab**
ConfigMgr CB 2509 · Build 5.00.9141 · SQL Server 2022 · Hyper-V Lab

June 2026 · Internal Lab Documentation

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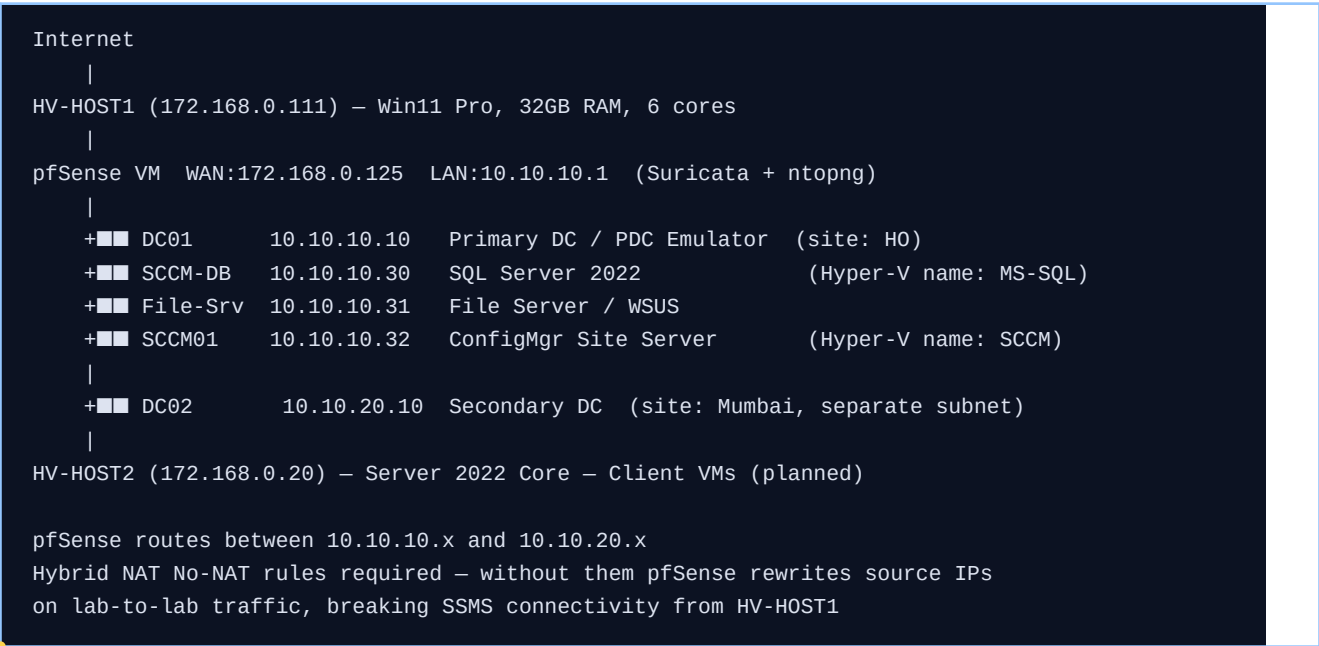
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1. Lab Environment & Architecture

A dual-host Hyper-V lab built for ConfigMgr endpoint management skills, with a secondary goal of routing SCCM content through pfSense for Suricata/ntopng IDS testing. All production VMs run on HV-HOST1 (Win11 Pro). HV-HOST2 is reserved for client VMs (deployment pending).

Network Topology

Lab Network Diagram



■ Critical Naming Note — Hyper-V Display Name vs Guest Hostname

Hyper-V display names (used in Get-VM*, Disable-VMIntegrationService etc.) differ from in-guest hostnames used in DNS, PowerShell remoting, and all other commands. Display name "SCCM" = guest hostname SCCM01 (10.10.10.32) Display name "MS-SQL" = guest hostname SCCM-DB (10.10.10.30) Mixing these caused repeated "VM not found" errors throughout this build. Always verify which context you are operating in before running a command.

VM Inventory

Hyper-V Display Name	Guest Hostname	IP	Role	Host
pfSense	pfsense	10.10.10.1 / 172.168.0.125	Firewall/Router, Suricata, ntopng	HV-HOST1
DC01	DC01	10.10.10.10	Primary DC, PDC Emulator, DNS	HV-HOST1
DC02	DC02	10.10.20.10	Secondary DC (Mumbai site)	HV-HOST1

MS-SQL	SCCM-DB	10.10.10.30	SQL Server 2022 (Default instance)	HV-HOST1
File-Srv	File-Srv	10.10.10.31	File Server, WSUS (WID)	HV-HOST1
SCCM	SCCM01	10.10.10.32	ConfigMgr Site Server	HV-HOST1

Software & Account Reference

Item	Value / Detail
ConfigMgr Version	Current Branch 2509, Build 5.00.9141, Evaluation 360 days
Site Code / Name	LAB / Corp Lab Primary Site
SQL Instance	MSSQLSERVER (Default), Port 1433, Broker Port 4022
SQL Collation	SQL_Latin1_General_CP1_CI_AS (mandatory for ConfigMgr)
SQL Database	CM_LAB
SQL Data Drive	E:\ — NVMe, 40 GB, 64 KB allocation — CM_LAB.mdf
SQL Log Drive	F:\ — 15 GB — CM_LAB_log.ldf
SQL Max Memory	6,144 MB (capped; leave headroom for OS)
svc_sccm	SCCM service account — SQL sysadmin + securityadmin
svc_naa	Network Access Account
corpadmin	Primary admin — SQL sysadmin, Domain Admin, ConfigMgr Full Admin
CORP\SCCM01\$	Site server computer account — SQL sysadmin, AD delegated (Full Control on System Management)
WSUS	Installed on File-Srv using Windows Internal Database (WID)
AD Domain	corp.lab — Sites: HO (10.10.10.x), Mumbai (10.10.20.x)

2. Pre-Installation Preparation

Before running the SCCM setup wizard, several infrastructure components must be in place. This section covers what was configured and the non-obvious pitfalls discovered during preparation.

2.1 SQL Server 2022 Configuration

Key non-default settings required for ConfigMgr — verify all before running setup:

SQL — verification queries (run via sqlcmd or SSMS on SCCM-DB)

```
-- 1. Confirm collation (MUST be SQL_Latin1_General_CP1_CI_AS)
SELECT SERVERPROPERTY('Collation')

-- 2. Confirm Service Broker enabled on CM_LAB
-- (ConfigMgr uses SSB on port 4022 for database notification)
SELECT name, is_broker_enabled FROM sys.databases WHERE name = 'CM_LAB'

-- 3. Confirm max memory cap
EXEC sp_configure 'max server memory (MB)'
-- run_value should be 6144 (not 0 = unlimited)

-- 4. Confirm SCCM01$ has sysadmin
SELECT sp.name, r.name AS Role
FROM sys.server_role_members rm
JOIN sys.server_principals r ON rm.role_principal_id=r.principal_id
JOIN sys.server_principals sp ON rm.member_principal_id=sp.principal_id
WHERE sp.name LIKE '%SCCM01%'
-- Expected: CORP\SCCM01$ | sysadmin
--           CORP\SCCM01$ | securityadmin
```

→ Accessing SSMS from HV-HOST1 (Non-Domain-Joined)

HV-HOST1 (Win11) is NOT domain-joined. To connect SSMS to SCCM-DB: runas /netonly /user:CORP\corpadmin "ssms.exe" Use NETBIOS prefix (CORP\) not FQDN (corp.lab\). Also: MSOLEDBSQL 19.4 requires BOTH vc_redist.x64.exe AND vc_redist.x86.exe installed on the host — even on a 64-bit OS. Missing either causes a silent install failure.

2.2 Active Directory Preparation

Step 1: Extend the AD Schema

Run from the ConfigMgr installation media on a machine with Schema Admin rights:

CMD — DC01 (Schema Admin required)

```
# Run on DC01 as Schema Admin (or from SCCM01 if account has Schema Admin)
.\SMSSETUP\BIN\X64\extadsch.exe

# Verify – check C:\ExtADSch.log on the machine where it ran:
# "Successfully extended the Active Directory schema."

# Also verify the schema version in AD:
Get-ADObject "CN=MS-SMS-Site,CN=Schema,CN=Configuration,DC=corp,DC=lab" -Properties *
# Should exist if schema was extended successfully
```

Step 2: Create System Management Container and Delegate Permissions

This step is the source of Issue 3b (ACL inheritance gap). The correct delegation command is shown here to document what **should** have been done from the start:

PowerShell — DC01 or SCCM01 (with AD module)

```
# Create the container (skip if it already exists)
$de = [ADSI]"LDAP://CN=System,DC=corp,DC=lab"
$new = $de.Create("container","CN=System Management")
$new.SetInfo()

# Delegate – CORRECT: /I:T = this object AND all descendants
# Single quotes around trustee:rights – critical for PowerShell (see Learning #9)
dscls "CN=System Management,CN=System,DC=corp,DC=lab" /G 'CORP\SCCM01$:GA' /I:T

# Verify inheritance is complete
Import-Module ActiveDirectory
(Get-Acl "AD:\CN=System Management,CN=System,DC=corp,DC=lab").Access |
    Where-Object {$_.IdentityReference -like "*SCCM01*"} | Format-List

# Must show:
#   InheritanceType : All
#   InheritanceFlags : ContainerInherit, ObjectInherit <-- BOTH required
#   IsInherited      : False
```

✗ This Step Seeded Issue 3b

If the delegation is applied via the GUI or dscls without /I:T, InheritanceType will be None. SCCM setup succeeds (it runs as corpadmin = Domain Admin) but later SCCM01\$ cannot update child objects (SMS-Site-LAB, MP objects) because its ACE does not cover them. This causes persistent hman.log error code = 5 failures after installation. See Section 7 for the full diagnosis and fix.

2.3 Prerequisites Checklist

Prerequisite	Target	Status	Note
.NET Framework 4.5+	SCCM01	✓	Windows Features
Windows ADK + WinPE	SCCM01	✓	Required for OSD and boot images

WSUS role	File-Srv	✓	WID (Windows Internal Database) — sufficient for lab
SQL Server 2022	SCCM-DB	✓	Default instance, port 1433
SQL collation	SCCM-DB	✓	SQL_Latin1_General_CP1_CI_AS
SQL data/log drives	SCCM-DB	✓	E:\ NVMe 64KB alloc, F:\ logs
SCCM01\$ SQL sysadmin	SCCM-DB	✓	Added manually via SSMS
corpadmin SQL sysadmin	SCCM-DB	✓	Required to run setup wizard
AD Schema extended	DC01	✓	extadsch.exe from media
System Management container	DC01	■	Created but ACL misconfigured — see §7
MSOLEDBSQL 19.4	SCCM01	✓	Both x64 + x86 vc_redist required
SMB/WMI firewall rules	SCCM-DB	✗	MISSING — caused Issues 1 & 2

3. Issue 1 — WMI Firewall (Error 0x800706ba)

3.1 What Setup Does at This Step

When you enter the SQL data/log paths in the setup wizard, ConfigMgr does not do a simple file-system check. It connects to SCCM-DB's WMI namespace `root\cimv2` to query drives and validate the paths exist on the remote machine. This uses DCOM/RPC: first to port 135 (RPC Endpoint Mapper), then to a dynamic port assigned to the WMI provider host. Both hops must pass through Windows Firewall on SCCM-DB.

3.2 Symptom & Error

ConfigMgrSetupWizard.log — SCCM01 (Attempt 1)

```
Validating sqlDataPath = [\\SCCM-DB\E$\SQL_Data]
Validating sqlDataPath = [\\SCCM-DB\F$\SQL_Logs]

CWmi::Connect(): ConnectServer(Namespace) failed. - 0x800706ba
    Connecting to root\cimv2 on SCCM-DB.corp.lab
    0x800706ba = ERROR_RPC_S_SERVER_UNAVAILABLE
        "The RPC server is unavailable."
```

0x800706ba = ERROR_RPC_S_SERVER_UNAVAILABLE. The RPC infrastructure on SCCM-DB could not be reached. In the context of a WMI call this almost always means inbound WMI/RPC traffic is blocked by Windows Firewall on the remote machine.

3.3 Root Cause

Fresh Windows Server 2022 builds disable all non-essential inbound firewall rule groups. The "Windows Management Instrumentation (WMI)" group — which governs remote WMI access — was disabled on SCCM-DB.

Rule Name	Port/Protocol	Purpose
WMI-In (DCOM-In)	TCP 135	DCOM Endpoint Mapper — first contact point for all WMI calls
WMI-In (WMI-In)	TCP dynamic	WMI provider host — actual namespace queries
WMI-In (Async)	TCP dynamic	Asynchronous WMI event callbacks

■ Why This Is Easy to Miss

RDP (port 3389) has its own separate firewall rule that IS enabled by default. SQL Server port 1433 you opened manually. So the machine is reachable in every way you tested it — but any WMI-based remote query fails silently until this group is explicitly enabled. There is no warning during OS setup.

3.4 Fix & Verification

PowerShell — SCCM-DB (enable commands), SCCM01 (connectivity test)

```
# On SCCM-DB
Enable-NetFirewallRule -DisplayGroup "Windows Management Instrumentation (WMI)"
Enable-NetFirewallRule -DisplayGroup "File and Printer Sharing" # For E$ and F$ admin shares

# Verify on SCCM-DB
Get-NetFirewallRule -DisplayGroup "Windows Management Instrumentation (WMI)" |
    Select-Object DisplayName, Enabled

# Verify connectivity from SCCM01
Test-NetConnection SCCM-DB.corp.lab -Port 135
# Expected: TcpTestSucceeded : True
```

[SCREENSHOT PLACEHOLDER]

Session 1 — WMI firewall group on SCCM-DB, all rules Enabled: True after fix

Session 1 — WMI firewall group on SCCM-DB, all rules Enabled: True after fix

✓ Result

Setup progressed past UNC path validation. Fix is complete and correct for this issue.

4. Issue 2 — Remote Service Management Firewall (Win32 Error 258)

4.1 The Bootstrap Service Mechanism

To create the SQL certificate used for site-to-SQL encryption, setup on SCCM01 does the following against SCCM-DB:

1. Copies bootstrap files to a temp share on SCCM-DB (E:\SMS_SCCM01.corp.lab_SMS_SQL_SERVER<N>\)
2. Remotely installs service SMS_SERVER_BOOTSTRAP_SCCM01.corp.lab_SMS_SQL_SERVER via Windows SCM
3. Starts the service on SCCM-DB
4. **Waits for SCM to report service status back** via the Remote Service Management RPC channel
5. The service runs, creates the cert, writes to registry, stops itself — in under 150ms
6. Setup expects to observe Running → Stopped; if it never gets that notification it times out

■ Key Distinction

Steps 2-3 (install + start service) use the SCM over standard RPC — this worked fine. Step 4 (receive service status callback) uses the Remote Service Management firewall group — this was blocked. These are governed by TWO DIFFERENT firewall rule groups.

4.2 Symptom — Two Identical 59-Minute Failures

Setup failed at "Generating public key and SQL Server certificate" with **Win32 error 258 (WAIT_TIMEOUT)** after ~59 minutes, on two consecutive attempts. The identical timing was the critical signal: deterministic failures are not race conditions.

✗ Diagnostic Mistake #1 — Wrong Initial Call

"This is a race condition timing issue — the service exits before setup can observe the Running state. Just delete the stale service and retry." This was **WRONG**. Two reasons: (1) race conditions produce variable timing, not identical 59-minute timeouts; (2) even if setup missed the Running state, it would still detect Stopped eventually — a complete timeout indicates a blocking call that never returned, not a missed state transition. Should have checked firewall rule groups immediately.

4.3 srvboot.log: What Actually Happened on SCCM-DB

The ground truth is in `srvboot.log` on SCCM-DB, at:

E:\SMS_SCCM01.corp.lab_SMS_SQL_SERVER1\srvboot.log

srvboot.log — SCCM-DB (Attempt 2, 06-17-2026)

```
SMS_SERVER_BOOTSTRAP_SCCM01.corp.lab_SMS_SQL_SERVER started.
$$<SMS_BOOTSTRAP><06-17-2026 04:34:17.859+42>

Creating machine self-signed certificate for site role [SMS_SQL_SERVER]
  on Server [SCCM-DB]...
$$<SMS_BOOTSTRAP><06-17-2026 04:34:17.890+42>

Created machine certificate for site role [SMS_SQL_SERVER] successfully.
$$<SMS_BOOTSTRAP><06-17-2026 04:34:17.921+42> ← 62ms after start

Bootstrap operation successful.
SMS_SERVER_BOOTSTRAP_SCCM01.corp.lab_SMS_SQL_SERVER stopped.
$$<SMS_BOOTSTRAP><06-17-2026 04:34:17.937+42> ← 78ms total

Certificate created in: 62ms
Service ran for:      78ms
Setup gave up (SCCM01): 05:32:10 (~59 min 52 sec after service stopped)
```

The certificate was created in **62 milliseconds**. The service completed in **78 milliseconds**. The work succeeded. SCCM01 simply never received confirmation.

```
PS E:\SMS_SCCM01.corp.lab_SMS_SQL_SERVER1> Get-NetFirewallRule -DisplayGroup "Remote Service Management" | Select DisplayName, Enabled, Direction, Action
-----
DisplayName                Enabled Direction Action
-----
Remote Service Management (RPC)      False  Inbound  Allow
Remote Service Management (RPC-EPMAP) False  Inbound  Allow
Remote Service Management (NP-In)     False  Inbound  Allow

PS E:\SMS_SCCM01.corp.lab_SMS_SQL_SERVER1> Enable-NetFirewallRule -DisplayGroup "Remote Service Management"
PS E:\SMS_SCCM01.corp.lab_SMS_SQL_SERVER1> Get-NetFirewallRule -DisplayGroup "Remote Service Management" | Select DisplayName, Enabled, Direction, Action
-----
DisplayName                Enabled Direction Action
-----
Remote Service Management (RPC)      True   Inbound  Allow
Remote Service Management (RPC-EPMAP) True   Inbound  Allow
Remote Service Management (NP-In)     True   Inbound  Allow

PS E:\SMS_SCCM01.corp.lab_SMS_SQL_SERVER1> sc.exe delete SMS_SERVER_BOOTSTRAP_SCCM01.corp.lab_SMS_SQL_SERVER
[SC] DeleteService SUCCESS
PS E:\SMS_SCCM01.corp.lab_SMS_SQL_SERVER1>
```

Screenshot: Remote Service Management firewall group — all three rules (RPC, RPC-EPMAP, NP-In) showing Enabled: True after fix. sc.exe delete confirming stale service cleanup.

4.4 Why the First Diagnosis Was Wrong

Wrong Assumption	Why It Was Wrong
"Race condition — service exits too fast"	Race conditions produce variable timing. Identical 59-min timeouts = systematic block.
"Setup missed the Running state"	Even missing Running, setup would see Stopped. A full timeout = blocking call, not missed poll.
"The WMI fix from Issue 1 covered this"	WMI and Remote Service Management are separate firewall groups covering different RPC channels.

4.5 Root Cause

The "Remote Service Management" Windows Firewall rule group was disabled on SCCM-DB. This group governs remote SCM interactions — including service status notification callbacks from SCCM-DB back to SCCM01.

Firewall Rule Group	Governs	Status on SCCM-DB
Windows Management Instrumentation (WMI)	Remote WMI namespace queries	✗ Disabled → Fixed in Issue 1
File and Printer Sharing	Admin shares (E\$, F\$)	✗ Disabled → Fixed in Issue 1
Remote Service Management	SCM status callbacks	✗ Disabled → Root cause of Issue 2

4.6 Fix, Cleanup & Verification

PowerShell / CMD — SCCM-DB

```
# On SCCM-DB — check state before fix
Get-NetFirewallRule -DisplayGroup "Remote Service Management" |
  Select-Object DisplayName, Enabled, Direction
# All three show Enabled: False before fix

# Enable
Enable-NetFirewallRule -DisplayGroup "Remote Service Management"

# Verify — all three should show Enabled: True
Get-NetFirewallRule -DisplayGroup "Remote Service Management" |
  Select-Object DisplayName, Enabled

# Clean up stale stopped service from the failed attempt
# Run on SCCM-DB (service is already Stopped — no need to stop it first)
sc.exe delete SMS_SERVER_BOOTSTRAP_SCCM01.corp.lab_SMS_SQL_SERVER
# Expected: [SC] DeleteService SUCCESS
```

✓ Result

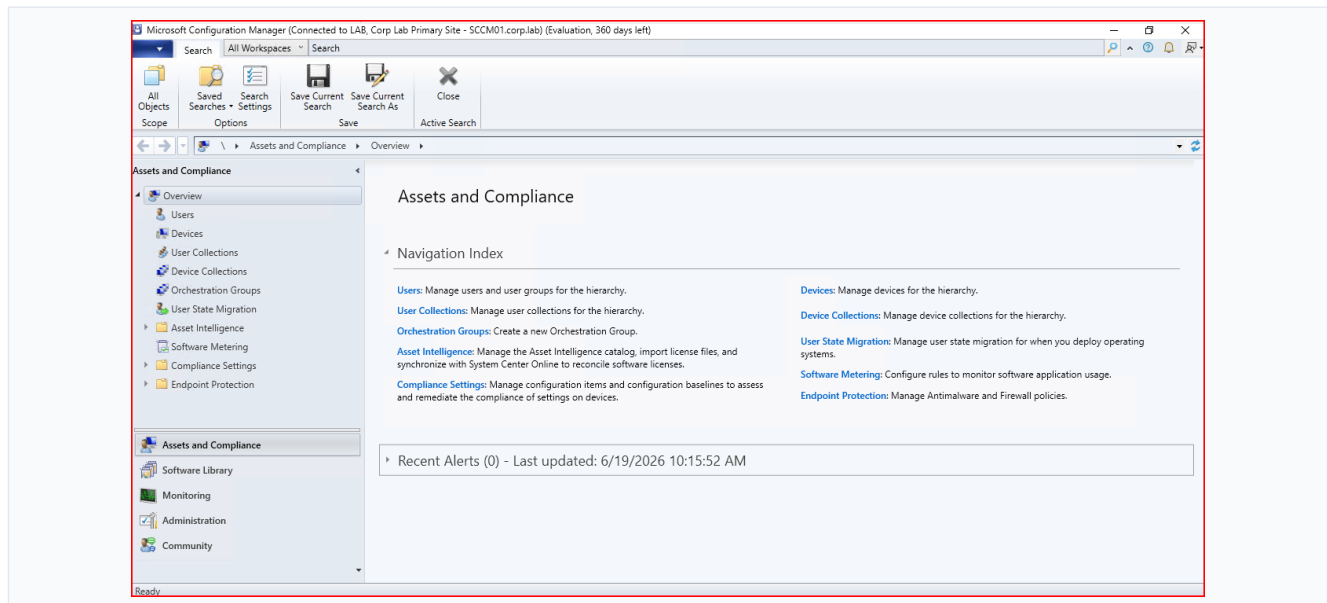
Third setup attempt succeeded. Certificate stage completed in 43 seconds (vs previous 59-minute timeouts). No other changes were required.

5. Site Installation — Success

5.1 Setup Log Stage Walkthrough

Time (6/18/2026)	Stage	Duration
05:06:31	Setup wizard launched	—
05:30:24	SQL connection established, paths auto-detected	<1 min
05:35:38	Manual path confirmed, page validated	1 min
05:35:49	Prerequisite checks started	2 min 45 sec
05:38:34	Installing ConfigMgr dependencies	1 min
05:38:35	Generating SQL Server certificate	43 sec (was 59 min)
05:39:18	Certificate done — removing old services/keys	<1 min
05:39:19	Setting up SQL Server database (CM_LAB)	13 min 10 sec
05:52:29	Copying files to SCCM01	5 min
05:57:28	Installing SMS Provider	13 sec
05:57:41	Installing registry	4 sec
05:57:45	Installing Site Component Manager	19 sec
05:58:04	Installing all services/components (parallel)	<1 min
05:58:06	Installing boot image packages	4 min 16 sec
06:02:22	Configuring data replication service	3 sec
06:02:25	Installing ConfigMgr console	31 sec
06:02:56	Creating program group	2 sec
06:02:58	SETUP HAS FINISHED	Total ~57 min

5.2 Console Connection Confirmed



Screenshot: ConfigMgr console connected — "Connected to LAB, Corp Lab Primary Site — SCCM01.corp.lab (Evaluation, 360 days left)". Recent Alerts: 0.

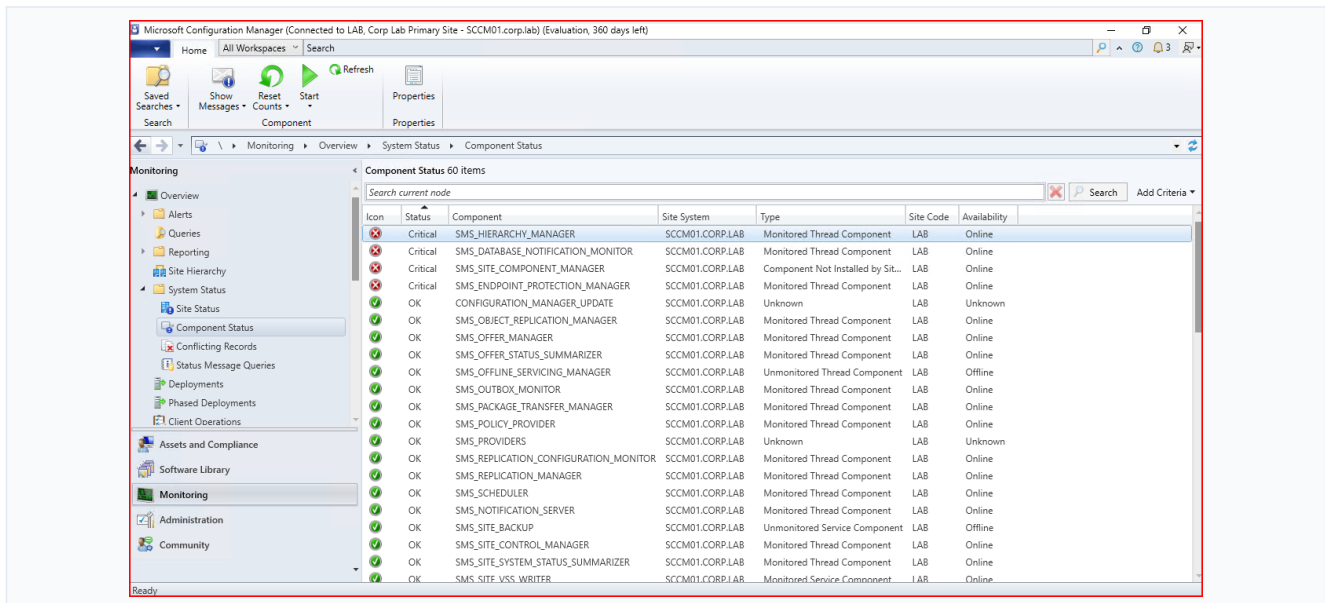
✓ Installation Complete

Console connects, site code LAB confirmed, evaluation license 360 days remaining, zero alerts. Issues 1 and 2 fully resolved at this point.

6. Issue 3a — Hyper-V Time Sync Conflict (Clock Skew)

6.1 Initial Component Status — Four Criticals

After installation, Component Status showed four Critical components. Last Status Message timestamps confirmed these were **live current failures**, not stale post-install noise.



Screenshot: Four Critical components. Note SMS_HIERARCHY_MANAGER has no Last Heartbeat — its thread fails before completing even one monitoring cycle.

Component Status 60 items								
Search current node								
Icon	Status	Component	Site System	Type	Site Code	Availability	Last Heartbeat	Last Status Message
✖	Critical	SMS_DATABASE_NOTIFICATION_MONITOR	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/19/2026 9:58 AM	6/19/2026 9:59 AM
✖	Critical	SMS_SITE_COMPONENT_MANAGER	SCCM01.CORP.LAB	Component Not Installed by Sit...	LAB	Online	6/19/2026 10:18 AM	6/19/2026 10:18 AM
✖	Critical	SMS_HIERARCHY_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/19/2026 10:18 AM
✖	Critical	SMS_ENDPOINT_PROTECTION_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/19/2026 10:18 AM	6/19/2026 9:59 AM

Screenshot: Timestamp detail confirms failures are from 6/19/2026 — the day after installation on 6/18, not stale install artifacts.

6.2 Ruling Out the Obvious

Check	Command	Result
SQL port 1433 reachable	Test-NetConnection SCCM-DB.corp.lab -Port 1433	✓ TcpTestSucceeded: True
SQL service running	Get-Service -Name MSSQLSERVER	✓ Running
SCCM01\$ SQL permissions	sqlcmd sys.server_role_members query	✓ sysadmin + securityadmin
Disk space E:\ (data)	Get-Volume -DriveLetter E,F	✓ 34.9 GB free / 40 GB

Disk space F:\ (logs)	Get-Volume -DriveLetter E,F	✓ 14.1 GB free / 15 GB
AD replication health	repadmin /showrepl	✓ All partitions successful
SCCM01\$ AD ACL	Get-Acl + dscls (System Management)	■ Appeared OK — bug found later (\$7)

6.3 Finding the Real Error in ddm.log

■ Log Filename Correction

ddm.log does NOT belong to SMS_DATABASE_NOTIFICATION_MONITOR despite what the name suggests. It belongs to SMS_DISCOVERY_DATA_MANAGER — confirmed by the SMS_DISCOVERY_DATA_MANAGER tag on every line. The correct log for SMS_DATABASE_NOTIFICATION_MONITOR was never identified — dbnotifymon.log does not exist in the Logs folder.

ddm.log — SCCM01 (06-20-2026 14:52 onwards, repeating every 60s)

```
Refreshing site settings.....
$$<SMS_DISCOVERY_DATA_MANAGER><06-20-2026 14:53:18.219-330>

*** [HY000][-2146893020][Microsoft][ODBC Driver 18 for SQL Server]
    SQL Server Network Interfaces: The clocks on the client and server
    machines are skewed.
$$<SMS_DISCOVERY_DATA_MANAGER><06-20-2026 14:53:18.223-330>

*** [HY000][-2146893020][Microsoft][ODBC Driver 18 for SQL Server]
    Cannot generate SSPI context
$$<SMS_DISCOVERY_DATA_MANAGER><06-20-2026 14:53:18.224-330>

*** Failed to connect to the SQL Server, connection type: SMS ACCESS.
$$<SMS_DISCOVERY_DATA_MANAGER><06-20-2026 14:53:18.226-330>

CDiscoverDataManager::Process - Could not refresh registry or site control
    values. Sleeping for 60 seconds~
$$<SMS_DISCOVERY_DATA_MANAGER><06-20-2026 14:53:18.206-330>

[Block repeats at 14:54:18, 14:55:18... every 60 seconds without stopping]
```

Name	Date modified	Type	Size
adctrl	6/20/2026 2:50 PM	Text Document	41 KB
AdminService	6/20/2026 2:50 PM	Text Document	437 KB
ADService	6/19/2026 9:58 AM	Text Document	2 KB
aikbmgr	6/20/2026 2:49 PM	Text Document	108 KB
bgbisapiMSI	6/18/2026 6:39 PM	Text Document	277 KB
bgbimgr	6/20/2026 2:53 PM	Text Document	984 KB
BgbServer	6/20/2026 2:53 PM	Text Document	437 KB
BgbSetup	6/18/2026 6:39 PM	Text Document	6 KB
BusinessAppProcessWorker	6/19/2026 9:58 AM	Text Document	4 KB
ccm	6/20/2026 2:41 PM	Text Document	259 KB
CertMgr	6/20/2026 2:50 PM	Text Document	559 KB
chmgr	6/19/2026 9:58 AM	Text Document	3 KB
CloudMgr	6/20/2026 2:53 PM	Text Document	290 KB
CmUnloadWmiProviders	6/18/2026 6:30 PM	Text Document	5 KB
CMUpdate	6/20/2026 2:48 PM	Text Document	211 KB
collevall	6/20/2026 2:53 PM	Text Document	1,448 KB
compmon	6/20/2026 2:50 PM	Text Document	286 KB
compsum	6/20/2026 2:53 PM	Text Document	2,148 KB
ComRegSetup	6/18/2026 6:28 PM	Text Document	1 KB
dataldr	6/20/2026 2:35 PM	Text Document	148 KB
ddm	6/20/2026 2:53 PM	Text Document	847 KB
despool	6/20/2026 2:50 PM	Text Document	47 KB
distmgr	6/20/2026 2:53 PM	Text Document	283 KB
EPMgr	6/20/2026 2:53 PM	Text Document	936 KB
FailOverMgr	6/20/2026 2:53 PM	Text Document	1,492 KB
hman	6/20/2026 2:53 PM	Text Document	1,150 KB
inboxmgr	6/20/2026 2:20 PM	Text Document	21 KB
inboxmon	6/20/2026 2:45 PM	Text Document	567 KB

Screenshot: ConfigMgr Logs folder — confirms dbnotifymon.log does not exist. ddm.log last modified 6/20/2026 2:53 PM (active).

6.4 Clock Skew and Kerberos — Why Everything Fails at Once

ODBC Driver 18 uses Windows Authentication (Kerberos) by default for domain-joined SQL connections. Kerberos requires all parties to agree on the current time within **5 minutes (the domain default tolerance)**. When SCCM01 and SCCM-DB clocks diverge beyond this, the KDC rejects Kerberos tickets as potentially replayed — surfacing as "Cannot generate SSPI context."

What Clock Skew Breaks	Mechanism
SQL Windows Authentication	Kerberos ticket rejected — timestamp out of tolerance
Active Directory LDAP writes	Same Kerberos failure for LDAP bind operations
ConfigMgr console (WMI/DCOM)	SMS Provider auth uses Kerberos — console cannot connect
Any SSPI-authenticated call	SSPI is Windows' interface to Kerberos — all affected simultaneously

■ Why Multiple Unrelated Components Failed at the Same Time

When SMS_HIERARCHY_MANAGER (AD writes), SMS_DATABASE_NOTIFICATION_MONITOR (SQL), SMS_SITE_COMPONENT_MANAGER (SQL), and SMS_ENDPOINT_PROTECTION_MANAGER all fail at the same timestamp — that is a Kerberos/clock problem, not four independent bugs. A single root cause (clock skew) broke all Windows-authenticated connections simultaneously.

6.5 Root Cause: Hyper-V Time Synchronization

Hyper-V's **Time Synchronization integration service** was enabled on both VMs (SCCM and MS-SQL). This service periodically pushes the Hyper-V host clock directly into the VM, competing with the domain w32time hierarchy.

Domain-joined Windows VMs should sync time exclusively via the domain hierarchy (NT5DS mode: member servers → DCs → PDC Emulator). When HV-HOST1 rebooted for Windows Updates at ~09:58 AM, the integration service stamped a diverged timestamp into both VMs, pushing their clocks beyond Kerberos tolerance relative to DC01/DC02.

PowerShell — HV-HOST1 (Win11 Hyper-V host)

```
# On HV-HOST1 (Win11 physical machine) — use Hyper-V DISPLAY names
Get-VMIntegrationService -VMName "SCCM","MS-SQL" |
    Where-Object {$_.Name -eq "Time Synchronization"}

# Output — both were Enabled: True (the problem)
VMName  Name                Enabled
-----  -
MS-SQL  Time Synchronization  True
SCCM    Time Synchronization  True
```

6.6 Fix Applied

Step 1: Disable Hyper-V Time Synchronization on both VMs:

PowerShell — HV-HOST1

```
# On HV-HOST1 — disable on both VMs
Disable-VMIntegrationService -VMName "SCCM","MS-SQL" -Name "Time Synchronization"

# Verify — both should show Enabled: False
Get-VMIntegrationService -VMName "SCCM","MS-SQL" |
    Where-Object {$_.Name -eq "Time Synchronization"}

VMName  Name                Enabled
-----  -
MS-SQL  Time Synchronization  False ✓
SCCM    Time Synchronization  False ✓
```

Step 2: On EACH VM, reconfigure w32time for domain hierarchy:

PowerShell — SCCM01 and SCCM-DB (run identically on each)

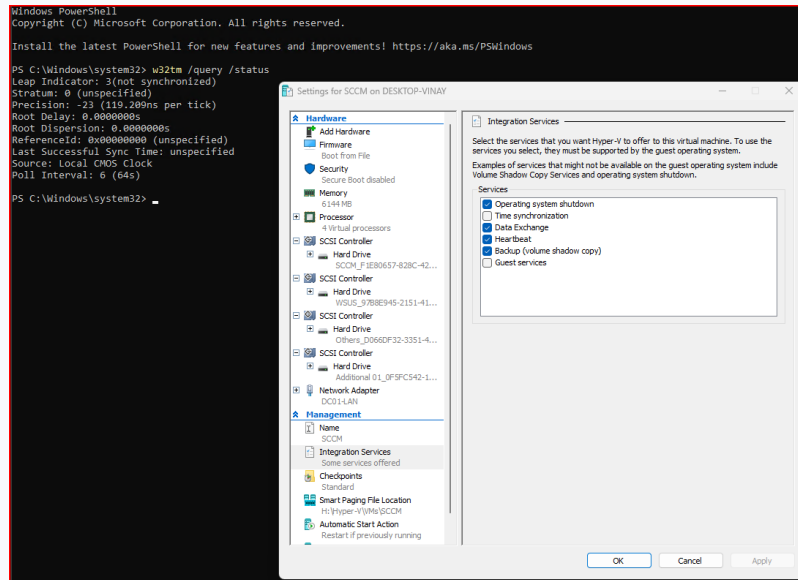
```
# Run on BOTH SCCM01 and SCCM-DB
w32tm /config /syncfromflags:domhier /update
Restart-Service w32time
w32tm /resync /rediscover

# Verify — expect Source: DC01.corp.lab, Stratum: 2
w32tm /query /status
```

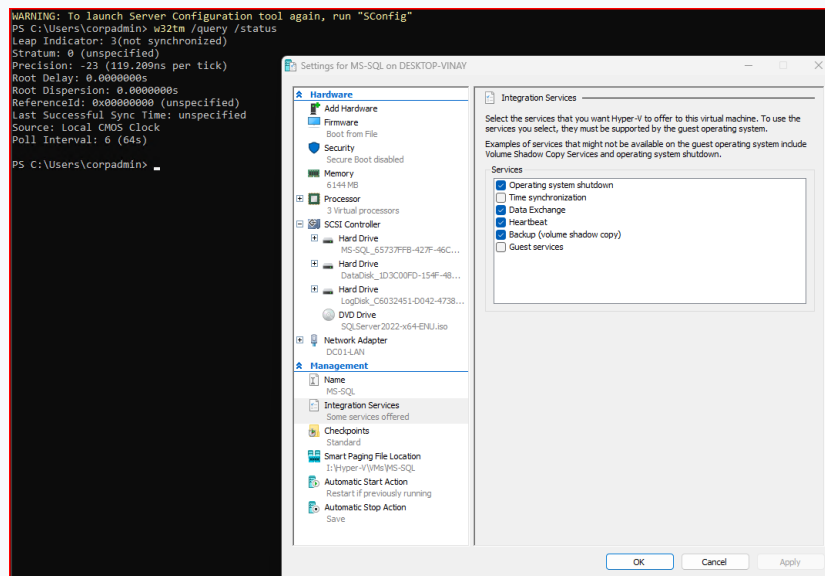
6.7 Regression: Stratum 0 Fallback

X What Went Wrong

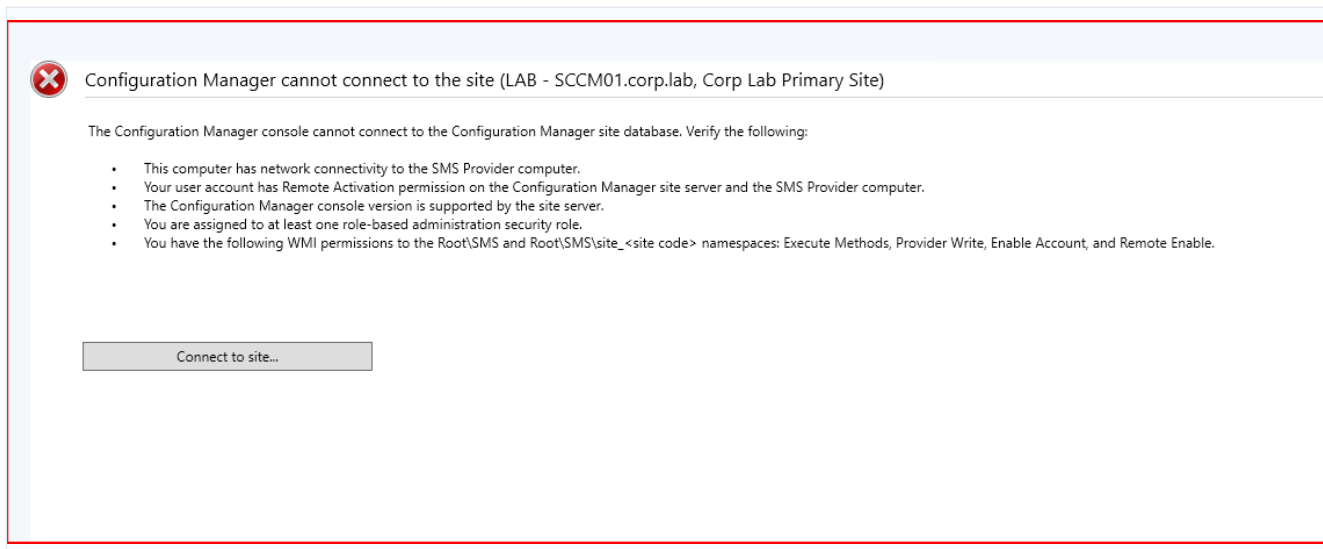
Running `w32tm /resync /force` immediately after disabling the integration service — before reconfiguring `w32time` with `/syncfromflags:domhier` — caused BOTH VMs to fall to Stratum 0 (Source: Local CMOS Clock, Leap Indicator: 3 not synchronized). Without the integration service AND without a configured domain source, `w32time` had no valid time source and fell back to the local hardware clock. This completely broke Kerberos — the ConfigMgr console could no longer connect at all.



Screenshot: SCCM01 — Stratum 0, Source: Local CMOS Clock after wrong resync sequence. Hyper-V settings panel confirms Time Synchronization is unchecked.



Screenshot: SCCM-DB — same Stratum 0 regression. Both VMs simultaneously off the domain time hierarchy.



Screenshot: ConfigMgr console "cannot connect to the site database" — Kerberos completely broken when both VMs have unsynced clocks.

6.8 Recovery Sequence

PowerShell — Recovery (SCCM01, then SCCM-DB)

```
# SCCM01 – /force worked here after domhier reconfiguration
w32tm /config /syncfromflags:domhier /update
Restart-Service w32time
w32tm /resync /force
w32tm /query /status
# Source: DC01.corp.lab Stratum: 2 ✓

# SCCM-DB – /force returned "no time data was available", use /rediscover
w32tm /config /syncfromflags:domhier /update
Restart-Service w32time
w32tm /resync /rediscover
w32tm /query /status
# Source: DC01.corp.lab Stratum: 2 ✓
```

→ /force vs /rediscover

/resync /force: use when w32time already knows its domain sources (running in domhier mode and sources are cached). Forces immediate sync. /resync /rediscover: use after a service restart or major config change. Forces w32time to re-query AD/DNS for time sources before syncing. If /force returns "no time data was available" — switch to /rediscover.

7. Issue 3b — AD System Management ACL Inheritance Gap

7.1 AD Write Failures in hman.log

After clock skew was resolved, three components remained in Warning/Critical. Filtering hman.log for errors revealed the actual failure:

PowerShell — SCCM01

```
# Filter hman.log for error lines
Select-String -Path "C:\Program Files\Microsoft Configuration Manager\Logs\hman.log"
-Pattern "error","fail","cannot","exception" -SimpleMatch |
Select-Object -Last 30
```

hman.log — SCCM01 (recurring error)

```
SMS-Site-LAB could not be updated, error code = 5.
$$<SMS_HIERARCHY_MANAGER><06-19-2026 09:58:53.591-330>

SMS-Site-LAB could not be updated, error code = 5.
$$<SMS_HIERARCHY_MANAGER><06-21-2026 09:15:03.120-330>

SMS-Site-LAB could not be updated, error code = 5.
$$<SMS_HIERARCHY_MANAGER><06-21-2026 10:05:12.483-330>

[Recurring approximately every 10-15 minutes across multiple days]

Configuration Manager cannot update the already existing object
"SMS-Site-LAB" in Active Directory (corp.lab).
```

Error code 5 = Windows ERROR_ACCESS_DENIED. The "already existing" qualifier is important — the object exists but SCCM01\$ cannot update it. This is an ACL issue, not a missing object or schema problem.

ⓧ	Milestone	LAB	6/21/2026 11:00:39 PM	SCCM01.CORP.LAB	SMS_HIERARCHY_MANAGER	4912	Configuration Manager cannot update the already existing object "SMS-Site-LAB" in Active Directory (corp.lab).	Possible cause: The site server's machine account may not have full control rights for the "System Management" cont
ⓧ	Milestone	LAB	6/21/2026 9:56:08 PM	SCCM01.CORP.LAB	SMS_HIERARCHY_MANAGER	4912	Configuration Manager cannot update the already existing object "SMS-Site-LAB" in Active Directory (corp.lab).	Possible cause: The site server's machine account may not have full control rights for the "System Management" cont
ⓧ	Milestone	LAB	6/21/2026 1:09:22 PM	SCCM01.corp.lab	SMS_HIERARCHY_MANAGER	4912	Configuration Manager cannot update the already existing object "SMS-Site-LAB" in Active Directory (corp.lab).	Possible cause: The site server's machine account may not have full control rights for the "System Management" cont
ⓧ	Milestone	LAB	6/21/2026 12:08:45 PM	SCCM01.corp.lab	SMS_HIERARCHY_MANAGER	4912	Configuration Manager cannot update the already existing object "SMS-Site-LAB" in Active Directory (corp.lab).	Possible cause: The site server's machine account may not have full control rights for the "System Management" cont
ⓧ	Milestone	LAB	6/21/2026 11:08:05 AM	SCCM01.corp.lab	SMS_HIERARCHY_MANAGER	4912	Configuration Manager cannot update the already existing object "SMS-Site-LAB" in Active Directory (corp.lab).	Possible cause: The site server's machine account may not have full control rights for the "System Management" cont
ⓧ	Milestone	LAB	6/21/2026 9:15:03 AM	SCCM01.CORP.LAB	SMS_HIERARCHY_MANAGER	4912	Configuration Manager cannot update the already existing object "SMS-Site-LAB" in Active Directory (corp.lab).	Possible cause: The site server's machine account may not have full control rights for the "System Management" cont
ⓧ	Milestone	LAB	6/20/2026 11:37:32 PM	SCCM01.CORP.LAB	SMS_HIERARCHY_MANAGER	4912	Configuration Manager cannot update the already existing object "SMS-Site-LAB" in Active Directory (corp.lab).	Possible cause: The site server's machine account may not have full control rights for the "System Management" cont
ⓧ	Milestone	LAB	6/20/2026 11:32:19 PM	SCCM01.corp.lab	SMS_HIERARCHY_MANAGER	3303	Hierarchy Manager cannot connect to the site database.	Possible cause: SQL Server problem. Solution: 1. Review the immediately preceding status messages from this component about SQL Server errors. 2. Verify that this comp
ⓧ	Milestone	LAB	6/20/2026 11:27:19 PM	SCCM01.corp.lab	SMS_HIERARCHY_MANAGER	3303	Hierarchy Manager cannot connect to the site database.	Possible cause: SQL Server problem. Solution: 1. Review the immediately preceding status messages from this component about SQL Server errors. 2. Verify that this comp
ⓧ	Milestone	LAB	6/20/2026 11:22:19 PM	SCCM01.corp.lab	SMS_HIERARCHY_MANAGER	3303	Hierarchy Manager cannot connect to the site database.	Possible cause: SQL Server problem. Solution: 1. Review the immediately preceding status messages from this component about SQL Server errors. 2. Verify that this comp
ⓧ	Milestone	LAB	6/20/2026 11:17:19 PM	SCCM01.corp.lab	SMS_HIERARCHY_MANAGER	3303	Hierarchy Manager cannot connect to the site database.	Possible cause: SQL Server problem. Solution: 1. Review the immediately preceding status messages from this component about SQL Server errors. 2. Verify that this comp
ⓧ	Milestone	LAB	6/20/2026 11:12:19 PM	SCCM01.corp.lab	SMS_HIERARCHY_MANAGER	3303	Hierarchy Manager cannot connect to the site database.	Possible cause: SQL Server problem. Solution: 1. Review the immediately preceding status messages from this component about SQL Server errors. 2. Verify that this comp
ⓧ	Milestone	LAB	6/20/2026 11:07:19 PM	SCCM01.corp.lab	SMS_HIERARCHY_MANAGER	3303	Hierarchy Manager cannot connect to the site database.	Possible cause: SQL Server problem. Solution: 1. Review the immediately preceding status messages from this component about SQL Server errors. 2. Verify that this comp

Screenshot: Hierarchy Manager status messages — recurring error code = 5 for SMS-Site-LAB across multiple timestamps, alongside earlier SQL errors from the clock skew period.

7.2 AD ACL Inheritance — How It Works

An ACE (Access Control Entry) in Active Directory has an **InheritanceType** property that controls exactly which objects the permission covers. Setting "Full Control" without the right inheritance flags silently creates a permission that looks correct but does nothing for child objects.

InheritanceType / Flags	Container Itself	Child Containers	Child Leaf Objects
-------------------------	---------------------	------------------	--------------------

None / None (what we had)	✓	✗	✗
All / ContainerInherit only (PS .All)	✓	✓	■ only if container-class
Subtree / ContainerInherit+ObjectInherit (dscls /I:T)	✓	✓	✓

■ The Correct Delegation for System Management

ConfigMgr creates objects of different AD schema classes inside this container (mSSMSSite = container-class, mSSMSManagementPoint = likely leaf-class). You need BOTH ContainerInherit AND ObjectInherit to guarantee coverage regardless of schema class. The only reliable way is dscls with /I:T (subtree).

7.3 Confirming the Misconfiguration

PowerShell — SCCM01

```
Import-Module ActiveDirectory
(Get-Acl "AD:\CN=System Management,CN=System,DC=corp,DC=lab").Access |
  Where-Object {$_.IdentityReference -like "*SCCM01*"} | Format-List
```

Get-Acl output — before fix

```
# Output BEFORE fix
ActiveDirectoryRights : GenericAll
InheritanceType       : None          ← Only covers the container itself
ObjectType            : 00000000-0000-0000-0000-000000000000
InheritedObjectType   : 00000000-0000-0000-0000-000000000000
ObjectFlags           : None
AccessControlType     : Allow
IdentityReference     : CORP\SCCM01$
IsInherited           : False
InheritanceFlags      : None          ← No ContainerInherit or ObjectInherit
PropagationFlags      : None
```

```
PS C:\Windows\system32> Import-Module ActiveDirectory
PS C:\Windows\system32>
PS C:\Windows\system32> $container = "CN=System Management,CN=System,DC=corp,DC=lab"
PS C:\Windows\system32>
PS C:\Windows\system32> (Get-Acl -Path "AD:\$container").Access | Where-Object {$_.IdentityReference -like "*SCCM01*"} | Format-List
ActiveDirectoryRights : GenericAll
InheritanceType       : None
ObjectType            : 00000000-0000-0000-0000-000000000000
InheritedObjectType   : 00000000-0000-0000-0000-000000000000
ObjectFlags           : None
AccessControlType     : Allow
IdentityReference     : CORP\SCCM01$
IsInherited           : False
InheritanceFlags      : None
PropagationFlags      : None

PS C:\Windows\system32> dscls "CN=System Management,CN=System,DC=corp,DC=lab" | findstr SCCM01
Allow CORP\SCCM01$          FULL CONTROL
PS C:\Windows\system32> _
```

Screenshot: Get-Acl confirming InheritanceType: None on the SCCM01\$ ACE. dscls shows "Allow CORP\SCCM01\$ FULL CONTROL" — appears correct but covers only the container, not child objects.

7.4 Fix Attempt 1 — PowerShell (Incomplete)

PowerShell — SCCM01 (one-liner format)

```
# Add new ACE with All inheritance
$Sid=(Get-ADComputer -Identity "SCCM01").SID
$acl=Get-Acl "AD:\CN=System Management,CN=System,DC=corp,DC=lab"
$acl.AddAccessRule((
    New-Object System.DirectoryServices.ActiveDirectoryAccessRule(
        $Sid,
        [System.DirectoryServices.ActiveDirectoryRights]::GenericAll,
        [System.Security.AccessControl.AccessControlType]::Allow,
        [System.DirectoryServices.ActiveDirectorySecurityInheritance]::All
    )))
Set-Acl "AD:\CN=System Management,CN=System,DC=corp,DC=lab" $acl
```

```
PS C:\Windows\system32> $Sid=(Get-ADComputer -Identity "SCCM01").SID; $acl=Get-Acl "AD:\CN=System Management,CN=System,DC=corp,DC=lab"; $acl.AddAccessRule((New-Object System.DirectoryServices.ActiveDirectoryAccessRule($Sid,[System.DirectoryServices.ActiveDirectoryRights]::GenericAll,[System.Security.AccessControl.AccessControlType]::Allow,[System.DirectoryServices.ActiveDirectorySecurityInheritance]::All))); Set-Acl "AD:\CN=System Management,CN=System,DC=corp,DC=lab" $acl
PS C:\Windows\system32> (Get-Acl "AD:\CN=System Management,CN=System,DC=corp,DC=lab").Access | Where-Object {$_.IdentityReference -like "*SCCM01*"} | Format-List
ActiveDirectoryRights : GenericAll
InheritanceType        : All
ObjectType             : 00000000-0000-0000-0000-000000000000
InheritedObjectType    : 00000000-0000-0000-0000-000000000000
ObjectFlags            : None
AccessControlType      : Allow
IdentityReference      : CORP\SCCM01$
IsInherited            : False
InheritanceFlags       : ContainerInherit
PropagationFlags       : None
```

Screenshot: Get-Acl after Attempt 1 — *InheritanceType* now "All", *InheritanceFlags* shows "ContainerInherit". Improvement but incomplete.

■ Partial Result

SMS_HIERARCHY_MANAGER cleared to OK — because SMS-Site-LAB is a container-class AD object, so ContainerInherit alone covered it. SMS_SITE_COMPONENT_MANAGER remained in Warning — cn=SMS-MP-LAB-SCCM01.CORP.LAB (the Management Point object) is a leaf-class object, which requires ObjectInherit — not provided by this fix.

7.5 Why Attempt 1 Was Incomplete

The PowerShell enum `ActiveDirectorySecurityInheritance.All` produced only **ContainerInherit** in the actual ACE flags. It did NOT set **ObjectInherit**:

ACE Flag	Covers	Set by .All enum?
ContainerInherit	Child objects that are container-class (can have children)	✓ Yes
ObjectInherit	Child objects that are leaf/non-container class	✗ No — this is the gap

✗ Bottom Line

Do not use the PowerShell `ActiveDirectoryAccessRule` approach for the System Management container delegation — the enum does not map cleanly to what ConfigMgr actually needs. Use `dscls /I:T` which sets both flags correctly and is explicit about the scope.

7.6 Fix Attempt 2 — dsacIs (Complete)

X PowerShell \$ Quoting — Do Not Skip This

The \$ in SCCM01\$ MUST be in single quotes when calling dsacIs from PowerShell. Double quotes cause PowerShell to attempt variable substitution on \$:GA, corrupting the argument silently and producing "The parameter is incorrect." WRONG: dsacIs "... " /G "CORP\SCCM01\$:GA" /I:T CORRECT: dsacIs "... " /G 'CORP\SCCM01\$:GA' /I:T

PowerShell / CMD — SCCM01

```
# /I:T = This object AND all descendant objects (full subtree)
# Single quotes around trustee:rights argument
dsacIs "CN=System Management,CN=System,DC=corp,DC=lab" /G 'CORP\SCCM01$:GA' /I:T

# Verify in the dsacIs output – look for this section:
# Permissions inherited to subobjects are:
# Inherited to all subobjects
# Allow CORP\SCCM01$    FULL CONTROL    ← This line must be present
```

7.7 Replication & Object Verification

PowerShell — SCCM01

```
# Force replication both directions – ensures both DCs have updated ACL
repadmin /syncall /AdeP
# Expected: "SyncAll terminated with no errors" for all 5 partitions

# Verify the MP object is in the correct location (not moved or orphaned)
Get-ADObject -LDAPFilter "(cn=SMS-MP-LAB-SCCM01.CORP.LAB)"
    -SearchBase "DC=corp,DC=lab" | Select-Object DistinguishedName
# Expected:
# CN=SMS-MP-LAB-SCCM01.CORP.LAB,CN=System Management,CN=System,DC=corp,DC=lab

# Verify inherited ACE is flowing down to the child object
$mp = "AD:\CN=SMS-MP-LAB-SCCM01.CORP.LAB,CN=System Management,CN=System,DC=corp,DC=lab"
(Get-Acl $mp).Access | Where-Object {$_.IdentityReference -like "*SCCM01*"}
# Key field: IsInherited : True
```

Component Status 60 items								
Search current node								
Icon	Status	Component	Site System	Type	Site Code	Availability	Last Heartbeat	Last Status Message
✓	OK	CONFIGURATION_MANAGER_UPDATE	SCCM01.CORP.LAB	Unknown	LAB	Unknown		
✓	OK	SMS_AI_KB_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:14 AM	6/21/2026 9:15 AM
✓	OK	SMS_ALERT_NOTIFICATION	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_BUSINESS_APP_PROCESS_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_CERTIFICATE_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:14 AM	6/21/2026 9:15 AM
✓	OK	SMS_CLIENT_CONFIG_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_CLIENT_HEALTH	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_CLOUD_SERVICES_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_COLLECTION_EVALUATOR	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_COMPONENT_MONITOR	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:15 AM	6/21/2026 9:15 AM
✓	OK	SMS_COMPONENT_STATUS_SUMMARIZER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:15 AM	6/21/2026 9:15 AM
⚠	Warning	SMS_DATABASE_NOTIFICATION_MONITOR	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:14 AM	6/21/2026 9:15 AM
✓	OK	SMS_DESPooler	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_DISCOVERY_DATA_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_DISTRIBUTION_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_EN_ADService_Monitor	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_ENDPOINT_PROTECTION_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:14 AM	6/21/2026 9:15 AM
✓	OK	SMS_EXECUTIVE	SCCM01.CORP.LAB	Monitored Service Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_FAILOVER_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:15 AM	6/21/2026 9:15 AM
⚠	Warning	SMS_HIERARCHY_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_INBOX_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:14 AM	6/21/2026 9:15 AM
✓	OK	SMS_INBOX_MONITOR	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:15 AM	6/21/2026 9:15 AM
✓	OK	SMS_INVENTORY_DATA_LOADER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_INVENTORY_PROCESSOR	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_LAN_SENDER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_LICENSE_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_MESSAGE_PROCESSING_ENGINE	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM

Screenshot: Component Status after all fixes — majority green. SMS_ENDPOINT_PROTECTION_MANAGER cleared from Critical to OK. SMS_DATABASE_NOTIFICATION_MONITOR and SMS_HIERARCHY_MANAGER now Warning.

Component Status 60 items								
Search current node								
Icon	Status	Component	Site System	Type	Site Code	Availability	Last Heartbeat	Last Status Message
✓	OK	SMS_MESSAGE_PROCESSING_ENGINE	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_MIGRATION_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_MP_CONTROL_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:14 AM	6/21/2026 9:15 AM
✓	OK	SMS_MP_FILE_DISPATCH_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:14 AM	6/21/2026 9:15 AM
✓	OK	SMS_NETWORK_DISCOVERY	SCCM01.CORP.LAB	Unmonitored Thread Component	LAB	Offline		6/21/2026 9:15 AM
✓	OK	SMS_NOTIFICATION_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_NOTIFICATION_SERVER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_OBJECT_REPLICATION_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_OFFER_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_OFFER_STATUS_SUMMARIZER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:15 AM	6/21/2026 9:15 AM
✓	OK	SMS_OFFLINE_SERVICING_MANAGER	SCCM01.CORP.LAB	Unmonitored Thread Component	LAB	Offline	6/21/2026 9:14 AM	6/21/2026 9:15 AM
✓	OK	SMS_OUTBOX_MONITOR	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:15 AM	6/21/2026 9:15 AM
✓	OK	SMS_PACKAGE_TRANSFER_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_POLICY_PROVIDER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_PROVIDERS	SCCM01.CORP.LAB	Unknown	LAB	Unknown		
✓	OK	SMS_REPLICATION_CONFIGURATION_MONITOR	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_REPLICATION_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_REST_PROVIDER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_RULE_ENGINE	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_SCHEDULER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_SITE_BACKUP	SCCM01.CORP.LAB	Unmonitored Service Component	LAB	Offline		6/21/2026 9:15 AM
⚠	Warning	SMS_SITE_COMPONENT_MANAGER	SCCM01.CORP.LAB	Component Not Installed by Sit...	LAB	Online	6/21/2026 9:15 AM	6/21/2026 9:15 AM
✓	OK	SMS_SITE_CONTROL_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_SITE_SQL_BACKUP_SCCM01.CORP.LAB	SCCM-DB.CORP.LAB	Monitored Service Component	LAB	Online	6/21/2026 9:05 AM	6/21/2026 9:15 AM
✓	OK	SMS_SITE_SYSTEM_STATUS_SUMMARIZER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:14 AM	6/21/2026 9:15 AM
✓	OK	SMS_SITE_VSS_WRITER	SCCM01.CORP.LAB	Monitored Service Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_SOFTWARE_INVENTORY_PROCESSOR	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM

Screenshot: Page 2 — SMS_SITE_COMPONENT_MANAGER in Warning (clearing). All other components OK.

✓	OK	SMS_SOFTWARE_METERING_PROCESSOR	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_STATE_SYSTEM	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_STATUS_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online		6/21/2026 9:15 AM
✓	OK	SMS_TEM	SCCM01.CORP.LAB	Unknown	LAB	Unknown		
✓	OK	SMS_WINNT_SERVER_DISCOVERY_AGENT	SCCM01.CORP.LAB	Unmonitored Thread Component	LAB	Online		6/21/2026 9:16 AM
✓	OK	SMS_WSUS_CONFIGURATION_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:15 AM	6/21/2026 9:15 AM
✓	OK	SMS_WSUS_SYNC_MANAGER	SCCM01.CORP.LAB	Monitored Thread Component	LAB	Online	6/21/2026 9:15 AM	6/21/2026 9:15 AM

Screenshot: Page 3 — SMS_WSUS_CONFIGURATION_MANAGER, SMS_WSUS_SYNC_MANAGER and all remaining components OK.

8. Diagnostic Methodology

8.1 Log File Reference

Log File	Component	What to Look For
ConfigMgrSetup.log	Setup Wizard	Full install log: stages, timings, errors
ConfigMgrSetupWizard.log	Setup GUI	Wizard-specific: SQL validation, prerequisite checks
srvboot.log	Bootstrap svc (on SCCM-DB!)	Certificate creation times. Path: E:\SMS_SCCM01...\SMS_SQL_SERVER\
hman.log	SMS_HIERARCHY_MANAGER	AD site publishing, SQL connect errors, topology
ddm.log	SMS_DISCOVERY_DATA_MANAGER	Clock skew / SSPI errors. NOT Database Notification Monitor.
sitecomp.log	SMS_SITE_COMPONENT_MANAGER	Component install/uninstall, AD object errors
EPMgr.log	SMS_ENDPOINT_PROTECTION_MANAGER	EP role status, signature errors
distmgr.log	SMS_DISTRIBUTION_MANAGER	Package/content distribution failures
mpcontrol.log	SMS_MP_CONTROL_MANAGER	Management Point health
CertMgr.log	SMS_CERTIFICATE_MANAGER	Certificate lifecycle
?? unknown ??	SMS_DATABASE_NOTIFICATION_MONITOR	Log filename NOT confirmed. dbnotifymon.log does not exist.

Log location: C:\Program Files\Microsoft Configuration Manager\Logs\

8.2 Step-by-Step Troubleshooting Approach

1

Confirm the failure is current, not stale

Check Last Status Message and Last Heartbeat timestamps in Component Status. A Critical badge from 24+ hours ago on a component that is now heartbeating normally is stale. Use Reset Counts (ribbon) to force re-evaluation.

2

Read the actual status message text first

Right-click → Show Messages → All. The console icon (red X) only tells you it is Critical. The message text gives you the actual error code. "Cannot connect to site database" and "error code = 5 ACCESS_DENIED" point to completely different root causes.

3**Check network and service basics — 30 seconds each**

Test-NetConnection for the relevant port. Get-Service for the relevant service. These rule out the most obvious causes before you invest time in log analysis.

4**Find and read the CORRECT log for the component**

The component name is not always the log file name. ddm.log = Discovery Data Manager, not Database Notification Monitor. Filter logs for errors: `Select-String -Pattern "error";"fail";"cannot";"exception"`

5**Check Windows Firewall rule groups on any new server first**

Before assuming permissions or network topology issues — verify all three groups on any fresh Windows Server: WMI, Remote Service Management, File and Printer Sharing. All three are disabled by default. Each produces a distinct failure.

6**When multiple components fail at the same timestamp — check clocks**

Simultaneous failures across unrelated components = platform-level event or clock skew. Run `w32tm /query /status` on all involved machines immediately. Look for Stratum: 0 or Source: Local CMOS Clock as the tell.

7**Check AD ACL inheritance flags, not just the presence of an ACE**

`dscls` showing "Full Control" and `Get-Acl` showing `GenericAll` are NOT sufficient. Explicitly check `InheritanceType` and `InheritanceFlags`. `InheritanceType: None` = permission only covers the container object itself.

9. Key Learnings

Non-obvious failure modes, diagnostic mistakes, and environment-specific gotchas discovered during this build. Each is specific enough to be actionable the next time it is encountered.

1**Fresh Windows Server Builds Default-Deny Three Required Firewall Groups [Firewall]**

WMI, Remote Service Management, and File/Printer Sharing are all disabled by default on fresh Windows Server 2022. Each produces a completely different error (0x800706ba RPC unavailable, Win32 258 WAIT_TIMEOUT) with no indication the firewall is the cause. Enable all three on any new server before running any ConfigMgr operation against it.

2**Deterministic Failures Are Not Race Conditions [Diagnosis]**

Two identical 59-minute timeouts = systematic block, not timing luck. Race conditions produce variable timing. When a failure reproduces identically on every attempt, the diagnosis should be "what is systematically preventing this?" not "just retry." The initial wrong call to retry without investigation cost one full extra setup attempt.

3**srvboot.log on SCCM-DB Is Ground Truth for Certificate Creation [Logs]**

ConfigMgrSetup.log on SCCM01 shows what setup THINKS happened. srvboot.log on SCCM-DB shows what ACTUALLY ran. In this case srvboot.log proved the certificate was created in 78ms — completely changing the diagnosis from "certificate creation failed" to "status notification was never received."

4**Disable Hyper-V Time Synchronization on ALL Domain-Joined VMs [Hyper-V]**

The Hyper-V Time Synchronization integration service competes with the domain w32time hierarchy. Domain-joined VMs should sync time exclusively via NT5DS mode (domain hierarchy from PDC Emulator). Disable the service on every domain-joined VM immediately when joining to the domain: `Disable-VMIntegrationService -VMName "" -Name "Time Synchronization"`

5**w32tm Resync Sequence: Disable → Reconfigure → Restart → /rediscover [Time Sync]**

Running `w32tm /resync /force` immediately after disabling Hyper-V time sync caused both VMs to fall to Stratum 0 (Local CMOS Clock). Correct sequence: (1) Disable integration service, (2) `w32tm /config /syncfromflags:domhier /update`, (3) `Restart-Service w32time`, (4) `w32tm /resync /rediscover`. Use `/rediscover` when `/force` returns "no time data was available."

6**Clock Skew Causes Simultaneous Failures Across All Kerberos-Authenticated Connections [Kerberos]**

When clocks diverge beyond 5 minutes, every Windows-authenticated connection fails at once: SQL ODBC, AD LDAP writes, WMI/DCOM. The signal is multiple unrelated components all failing at the same timestamp. Check clocks before assuming multiple independent bugs.

7

AD "Full Control" Is Silent if InheritanceType Is None [AD / ACL]

Setting GenericAll on the System Management container without inheritance grants rights only on the container object itself. Child objects (SMS-Site-LAB, Management Point objects) are not covered and SCCM01\$ gets ACCESS_DENIED (error 5). Always verify InheritanceFlags includes ContainerInherit AND ObjectInherit. Use dscls /I:T (subtree) for guaranteed full coverage.

8

PowerShell ActiveDirectorySecurityInheritance.All Does Not Set ObjectInherit [AD / ACL]

The .All enum value produces only ContainerInherit, not ObjectInherit. ContainerInherit covers container-class AD objects (SMS-Site-LAB). Leaf-class objects (Management Point, mSSMSManagementPoint) additionally need ObjectInherit. Do not use the PowerShell enum approach for this delegation — use dscls /I:T which sets both flags explicitly.

9

Single Quotes in PowerShell When \$ Appears in Arguments to External Programs

[PowerShell]

Double-quoting "CORP\SCCM01\$:GA" in PowerShell causes variable substitution on \$:GA, corrupting the argument before it reaches dscls. This surfaces as the misleading "The parameter is incorrect" error. Rule: any argument containing \$ passed to an external executable (dscls, sc.exe, etc.) must use single quotes in PowerShell.

10

Hyper-V Display Names Are Not Guest Hostnames [Hyper-V]

Hyper-V Manager display names differ from in-guest hostnames. Hyper-V host commands require display names ("SCCM", "MS-SQL"). All other commands use guest hostnames (SCCM01, SCCM-DB). Mixing these causes "VM not found" errors with no obvious indication of the naming mismatch. Confirm which context you are in before running.

11

ddm.log = Discovery Data Manager, NOT Database Notification Monitor [Logs]

Despite the filename, ddm.log is SMS_DISCOVERY_DATA_MANAGER. This was misidentified twice during diagnosis. The correct log for SMS_DATABASE_NOTIFICATION_MONITOR was not identified — dbnotifymon.log does not exist. This remains an open question for this lab.

12

pfSense Hybrid NAT No-NAT Rules Are Required for Lab-to-Lab Traffic [Network]

Without No-NAT rules, pfSense rewrites source IPs on inter-subnet lab traffic. This broke SSMS connectivity from HV-HOST1 to SCCM-DB early in the build. Any time cross-subnet connectivity works at the TCP level but authentication fails, check whether the source IP is being rewritten by NAT.

10. Command Quick Reference

Firewall — SCCM-DB

PowerShell — SCCM-DB (enable); SCCM01 (verify)

```
# Check and enable all three required groups
foreach ($g in @(
    "Windows Management Instrumentation (WMI)",
    "Remote Service Management",
    "File and Printer Sharing"
)) {
    Write-Host "=== $g ===" -ForegroundColor Cyan
    Get-NetFirewallRule -DisplayGroup $g | Select-Object DisplayName,Enabled
}

Enable-NetFirewallRule -DisplayGroup "Windows Management Instrumentation (WMI)"
Enable-NetFirewallRule -DisplayGroup "Remote Service Management"
Enable-NetFirewallRule -DisplayGroup "File and Printer Sharing"

# Verify connectivity from SCCM01 after enabling
Test-NetConnection SCCM-DB.corp.lab -Port 135 # RPC/WMI
Test-NetConnection SCCM-DB.corp.lab -Port 1433 # SQL
```

Bootstrap Service Cleanup — SCCM-DB

PowerShell/CMD — SCCM-DB

```
# List any leftover bootstrap services
Get-Service | Where-Object {$_.Name -like "SMS_SERVER_BOOTSTRAP*"}

# Delete (service must be Stopped state first)
sc.exe delete SMS_SERVER_BOOTSTRAP_SCCM01.corp.lab_SMS_SQL_SERVER
```

Hyper-V Time Synchronization — HV-HOST1

PowerShell — HV-HOST1 (Win11 Hyper-V host)

```
# Check state (use Hyper-V DISPLAY names, not guest hostnames)
Get-VMIntegrationService -VMName "SCCM","MS-SQL" |
    Where-Object {$_.Name -eq "Time Synchronization"}

# Disable on both VMs
Disable-VMIntegrationService -VMName "SCCM","MS-SQL" -Name "Time Synchronization"
```

w32time Domain Hierarchy Fix — SCCM01 and SCCM-DB

PowerShell — SCCM01 and SCCM-DB

```
# Run identically on BOTH machines
w32tm /config /syncfromflags:domhier /update
Restart-Service w32time
w32tm /resync /rediscover

# Verify
w32tm /query /status
# Success: Stratum: 2 Source: DC01.corp.lab
```

AD System Management Container ACL

PowerShell — SCCM01

```
# Correct delegation – single quotes around trustee:rights
dsaccls "CN=System Management,CN=System,DC=corp,DC=lab" /G 'CORP\SCCM01$:GA' /I:T

# Verify on container
Import-Module ActiveDirectory
(Get-Acl "AD:\CN=System Management,CN=System,DC=corp,DC=lab").Access |
  Where-Object {$_.IdentityReference -like "*SCCM01*"} | Format-List
# Must show InheritanceFlags: ContainerInherit, ObjectInherit

# Verify inherited ACE on child object
$mp="AD:\CN=SMS-MP-LAB-SCCM01.CORP.LAB,CN=System Management,CN=System,DC=corp,DC=lab"
(Get-Acl $mp).Access | Where-Object {$_.IdentityReference -like "*SCCM01*"}
# IsInherited must be True

# Force replication
repadmin /syncall /AdeP
```

SQL Verification

PowerShell — SCCM01 (for sqlcmd); SCCM-DB (for disk check)

```
# Check SCCM01$ role membership
sqlcmd -S SCCM-DB.corp.lab -E -Q "
  SELECT sp.name, r.name AS Role
  FROM sys.server_role_members rm
  JOIN sys.server_principals r ON rm.role_principal_id=r.principal_id
  JOIN sys.server_principals sp ON rm.member_principal_id=sp.principal_id
  WHERE sp.name LIKE '%SCCM01%'"

# Check disk space on SCCM-DB
Get-Volume -DriveLetter E,F |
  Select-Object DriveLetter,
    @{N="SizeGB";E={[math]::Round($_.Size/1GB,1)}},
    @{N="FreeGB";E={[math]::Round($_.SizeRemaining/1GB,1)}}
```

Log Analysis

PowerShell — SCCM01

```
$L = "C:\Program Files\Microsoft Configuration Manager\Logs"

# Tail a log
Get-Content -Tail 60 -Path "$L\hman.log"

# Filter for errors
Select-String -Path "$L\hman.log"
    -Pattern "error","fail","cannot","exception" -SimpleMatch |
    Select-Object -Last 30

# Watch in real time
Get-Content -Wait -Tail 20 -Path "$L\hman.log"
```

11. Issue Summary Table

#	Issue	Root Cause	Fix	Status
1	WMI 0x800706ba — UNC path validation failed	WMI firewall group disabled on SCCM-DB	Enable-NetFirewallRule "Windows Management Instrumentation (WMI)"	✓ Resolved
2	Win32 258 WAIT_TIMEOUT — bootstrap svc (×2 attempts)	Remote Service Management firewall group disabled on SCCM-DB	Enable-NetFirewallRule "Remote Service Management" + sc.exe delete stale svc	✓ Resolved
3a	Clock skew — Kerberos failures on SQL + AD (multiple components)	Hyper-V Time Sync integration service competing with domain w32time	Disable-VMIntegrationService + w32tm /config /syncfromflags:domhier	✓ Resolved
3a -R	Regression — Stratum 0 (Local CMOS Clock) on both VMs	w32tm /resync /force run before domhier reconfiguration	w32tm /config /syncfromflags:domhier + Restart-Service + /rediscover	✓ Resolved
3b	AD error 5 (ACCESS_DENIED) — SMS-Site-LAB + SMS-MP object	System Management ACE had InheritanceType: None — child objects excluded	dscls /G 'CORP\SCCM01\$:GA' /I:T + repadmin /syncall /AdeP	✓ Resolved

✓ Final Status — June 2026

All four originally-Critical components (SMS_HIERARCHY_MANAGER, SMS_DATABASE_NOTIFICATION_MONITOR, SMS_SITE_COMPONENT_MANAGER, SMS_ENDPOINT_PROTECTION_MANAGER) resolved or cleared from Critical status. ConfigMgr console connects successfully. Site LAB operational. Pending next steps: boundary/boundary group configuration, client VM deployment on HV-HOST2, pfSense Suricata/ntopng IDS testing once SCCM content traffic is flowing.